✦ EduGenie Application Brief

Gemini-Powered Learning Assistant & Interactive Educational Interface Technical Architecture Summary

Executive Summary

EduGenie is a web-based conversational application designed to provide interactive tutoring across multiple core academic disciplines. Powered by Google Gemini AI models, EduGenie simplifies complex subjects through customized explanations, practice problems, and diagnostic quizzes.

Core Capability Modules

SUBJECT MODULE

📐 Mathematics

Provides step-by-step solutions, theorem demonstrations (e.g., Pythagorean theorem), quadratic equations, and algebraic proofs with simple numerical examples.

SUBJECT MODULE

🔬 Science

Explains essential biological and physical processes like photosynthesis, thermodynamics, and planetary science using accessible analogies.

SUBJECT MODULE

📚 English & Grammar

Offers real-time grammar feedback, parts-of-speech identification, sentence construction exercises, and reading comprehension support.

SUBJECT MODULE

💻 Programming

Delivers interactive coding challenges, algorithm explanations, syntax assistance, and debugging hints for languages like JavaScript.

Interface Capabilities

- **Interactive Prompt Suggestions:** Pre-built quick action buttons to instantly initiate core learning workflows and subject overviews.
- **Adaptive Dark/Light Theme:** Full dynamic UI theme switching with persistent state saved in user local storage.
- **Responsive Layout Engine:** Mobile-first adaptive sidebar navigation and flexible message container sizing.
- **Rich Message Formatting:** Integrated formatting for bold highlights, code blocks, bullet points, and multi-line structured responses.

Application Component Matrix

The system architecture is organized into lightweight, high-performance UI components structured for seamless backend integration with the Gemini API.

Component	Responsibility	Technical Details
Sidebar Navigation	Quick subject selection, session management, and chat history tracking.	CSS Flexbox layout with responsive breakpoints for collapsible mobile view.
Chat Engine Container	Renders conversational thread, typing indicators, and user prompts.	Auto-scrolling dynamic DOM updates with animated typing indicators.
Input Bar	Multi-line user input handling with shortcut keys.	Auto-resizing textarea with keydown listener (`Enter` to submit, `Shift+Enter` for newline).
Theme Toggle	Switching between Light and Dark visual palettes.	Root CSS Variable mutation with `localStorage` key persistent storage.

Backend API Integration Strategy

To transition EduGenie from simulated demonstration mode to a full production assistant, standard integration with Google Cloud Gemini endpoints is configured as follows:

1. Authentication & Middleware

Route prompt requests through a secure Node.js/Python server layer to protect API keys and manage rate limits.

2. Contextual Prompt Design

Inject system instruction prefixes specifying pedagogical rules (e.g., "Act as an encouraging tutor, break down complex terms").

Production Readiness Roadmap

Key future upgrades include persistent conversation storage using PostgreSQL/MongoDB, multi-modal image analysis for homework troubleshooting, and voice-based question submission.